(5) A smart farming system predicts whether a crop is Healthy or Diseased based on environmental and plant conditions. The following training dataset contains 12 previously recorded samples.

S	Temp	Soil PH	Sunlight	Pest (?)	Leaf Condition	Crop Status
1	High	Acidic	Low	Yes	Yellow	Diseased
2	Moderate	Neutral	High	No	Green	Healthy
3	Low	Neutral	Mid	No	Green	Healthy
4	High	Acidic	Mid	Yes	Spotted	Diseased
5	Moderate	Alkaline	High	No	Green	Healthy
6	Low	Acidic	Low	Yes	Yellow	Diseased
7	Moderate	Neutral	High	No	Healthy	Healthy
8	High	Neutral	Mid	Yes	Spotted	Diseased
9	Low	Alkaline	High	No	Green	Healthy
10	Moderate	Acidic	Low	Yes	Yellow	Diseased
11	Low	Neutral	Mid	No	Healthy	Healthy
12	High	Alkaline	High	No	Green	Healthy

Given instance X = (Temp = Moderate, Soil PH = Neutral, Sunlight = High, Pest = No, Leaf condition = Green)

Using Naive Bayes Classifier, determine whether the crop is Healthy or Diseased.

Answer:

Given test instance,

X = (Temp = Moderate, Soil PH = Neutral, Sunlight = High, Pest = No, Leaf Condition = Green)

We need to determine if the crop is Healthy or Diseased.

Applying Naive Bayes in the given scenario:

$$P(C|X) \propto$$
$$P(C)\,P(x_1|C)\,P(x_2|C)\,P(x_3|C)\,P(x_4|C)\,P(x_5|C)$$

Where,

x_1 = Temp

x_2 = Soil PH

x_3 = Sunlight

x_4 = Pest

x_5 = Leaf Condition

We calculate this separately for

C = Healthy & C = Diseased

From the table,

Total Samples = 12

Healthy = 7

Diseased = 5

$\therefore P(\text{Healthy}) = \frac{7}{12}$

$\therefore P(\text{Diseased}) = \frac{5}{12}$

For $P(X \mid Healthy)$, we calculate the likelihood of each feature.

$\therefore P(Moderate \mid Healthy) = \frac{3}{7}$ // Temperature

(i) Among 7 healthy samples, 'Moderate' appears 3 times.

$\therefore P(Neutral \mid Healthy) = \frac{4}{7}$ // Soil PH

$\therefore P(High \mid Healthy) = \frac{5}{7}$ // Sunlight

$\therefore P(\cancel{Pest}\ No \mid Healthy) = \frac{7}{7} = 1$ // Pest

$\therefore P(Green \mid Healthy) = \frac{5}{7}$ // Leaf Condition

Calculating the Healthy score

$= \frac{7}{12} \times \frac{3}{7} \times \frac{4}{7} \times \frac{5}{7} \times 1 \times \frac{5}{7}$

$= 0.0728$

For $P(X \mid Diseased)$, we calculate the likelihood of each feature,

$\therefore P(Moderate \mid Diseased) = \frac{1}{5}$ // Temperature

$\therefore P(Neutral \mid Diseased) = \frac{1}{5}$ // Soil PH

$\therefore P(High \mid Diseased) = \frac{0}{5} = 0$ // Sunlight

$\therefore P(No \mid Diseased) = \frac{0}{5} = 0$ // Pest

$\therefore P(Green \mid Diseased) = \frac{0}{5} = 0$ // Leaf Condition

Calculating the Diseased score

$= \frac{5}{12}\left(\frac{1}{5} \times \frac{1}{5} \times 0 \times 0 \times 0\right) = 0$

We obtained,

Healthy score ≈ 0.0728

Diseased score $= 0$

Since, $0.0728 > 0$

The classifier predicts 'Healthy'.